

Practical 12: Customer Sales Dashboard Using Interactive Analytics

Aim

To design an interactive customer sales dashboard using a Superstore-style sales dataset, perform data cleaning and validation, calculate key performance indicators, analyze sales across regions, categories and time periods, create interactive visualizations, and interpret business insights for decision-making.

Problem definition

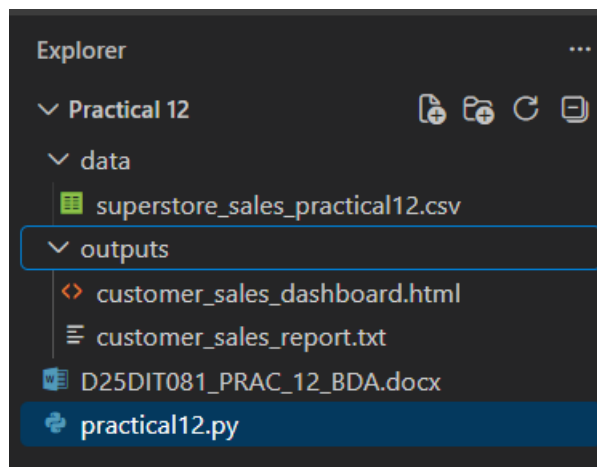
Management wants to monitor sales performance across products, regions, and customer segments through an interactive dashboard. Static reports are insufficient for decision-making. In this practical, a sales dataset is loaded, cleaned and analyzed to create KPIs, filters, charts, drill-down style analysis, summary cards and business insights.

Objectives

- Load and validate the Superstore sales dataset.
- Clean and prepare sales, profit, order and date fields.
- Create KPIs for Total Sales, Profit and Orders.
- Analyze sales using Region, Category and Time filters.
- Create interactive charts for sales and profitability.
- Perform product-wise profitability analysis.
- Perform customer segmentation analysis.
- Compare quarterly sales performance.
- Create an executive-style dashboard and interpret business insights.

Observation / Output (Screenshots)

Step 1: Creating Folder and files



Step 2: Loading Sales Data

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STEP 2: LOADING SALES DATA
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Dataset loaded successfully.
Records: 10000
Columns: ['Order_ID', 'Order_Date', 'Month', 'Region', 'State', 'City', 'Segment', 'Category', 'Sub_Category', 'Product_Name', 'Customer_Name', 'Quantity', 'Unit_Price', 'Discount', 'Sales', 'Profit']
Loading time: 0.05 seconds
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Step 3: Data Cleaning and Validation

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STEP 3: DATA CLEANING AND VALIDATION
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Records before cleaning: 10000
Records after cleaning: 10000
Duplicate rows removed: 0
Missing values: 0
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Step 4: Dataset Schema and Structure

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STEP 4: DATASET SCHEMA AND STRUCTURE
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Order_ID          str
Order_Date        datetime64[us]
Month             str
Region            str
State             str
City              str
Segment           str
Category          str
Sub_Category      str
Product_Name      str
Customer_Name     str
Quantity          int64
Unit_Price        float64
Discount          float64
Sales             float64
Profit            float64
Year              int32
Quarter           str
Month_Name        str
dtype: object

Sample records:
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Sample records:														
Order_ID	Order_Date	Month	Region	State	City	Segment	Category	Sub_Category	Product_Name	Customer_Name	Quantity	Unit_Price	Discount	Sales
ORD-00001	2025-11-23	2025-11	West	California	San Francisco	Consumer	Furniture	Chairs	Ergonomic Chair	Anaya Desai	2	564.20	0.00	1128.40
ORD-00002	2025-08-26	2025-08	East	Pennsylvania	Philadelphia	Consumer	Office Supplies	Paper	Printer Paper	Dev Joshi	5	122.37	0.00	611.85
ORD-00003	2025-01-24	2025-01	South	Virginia	Richmond	Consumer	Technology	Phones	Pixel 8	Vivaan Mehta	2	579.49	0.05	1158.98
ORD-00004	2025-08-14	2025-08	West	Arizona	Phoenix	Home Office	Furniture	Chairs	Visitor Chair	Ishita Shah	4	657.75	0.00	2631.00
ORD-00005	2024-06-15	2024-06	Central	Texas	Houston	Corporate	Technology	Laptops	HP Pavilion	Ishita Shah	6	574.75	0.20	3448.50

Step 5: KPI Calculation

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STEP 5: KPI CALCULATION
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Total Sales: $19,856,617.22
Total Profit: $2,721,446.80
Total Orders: 10,000
Total Quantity: 45,132
Profit Margin: 13.71%
Average Order Value: $1,985.66
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Step 6: Region Analysis

STEP 6: REGION ANALYSIS

Region	Sales	Profit	Orders	Quantity
South	5028526.04	694823.39	2505	11438
West	5000277.18	674197.13	2539	11503
Central	4948627.40	680478.97	2467	11187
East	4879186.60	671947.31	2489	11004

Step 7: Category Analysis

STEP 7: CATEGORY ANALYSIS

Category	Sales	Profit	Orders	Quantity
Technology	11316222.45	1757778.36	3314	14957
Furniture	7401822.73	778675.77	3378	15259
Office Supplies	1138572.04	184992.67	3308	14916

Step 8: Time Analysis

STEP 8: TIME ANALYSIS

Quarterly Sales:

Year	Quarter	Sales	Profit	Orders
2024	Q1	2457935.13	336639.26	1254
2024	Q2	2388566.02	334759.54	1232
2024	Q3	2506587.72	341980.93	1295
2024	Q4	2557392.66	355883.78	1267
2025	Q1	2533769.15	343108.00	1231
2025	Q2	2423678.67	330531.56	1225
2025	Q3	2474767.43	332907.97	1228
2025	Q4	2513920.44	345635.76	1268

Step 9: Product-wise Profitability

STEP 9: PRODUCT-WISE PROFITABILITY

Top 10 products by profit:

Category	Sub_Category	Product_Name	Sales	Profit	Quantity	Orders
Technology	Phones	Galaxy S24	1035674.64	162144.64	1307	302
Technology	Phones	Pixel 8	967124.96	158375.71	1266	269
Technology	Laptops	Dell Inspiron	1000039.85	154821.16	1367	291
Technology	Laptops	MacBook Air	940224.12	154021.51	1242	266
Technology	Accessories	USB-C Hub	1012422.74	152314.53	1350	295
Technology	Phones	iPhone 15	948127.73	149926.69	1211	285
Technology	Laptops	Lenovo ThinkPad	955370.93	145480.91	1230	272
Technology	Accessories	Webcam	915218.47	140165.60	1216	266
Technology	Laptops	HP Pavilion	911973.32	138961.81	1272	281
Technology	Accessories	Wireless Mouse	902563.57	136818.41	1179	260

Step 10: Customer Segmentation

STEP 10: CUSTOMER SEGMENTATION

Segment	Sales	Profit	Orders
Consumer	6649683.90	898898.95	3371
Home Office	6628130.28	920549.94	3296
Corporate	6578803.04	901997.91	3333

Step 11: Interactive Filters

STEP 11: INTERACTIVE FILTERS

Available Region filters: ['Central', 'East', 'South', 'West']
Available Category filters: ['Furniture', 'Office Supplies', 'Technology']
Available Segment filters: ['Consumer', 'Corporate', 'Home Office']
Interactive dashboard filters prepared successfully.

Step 12: Dashboard Generation

STEP 12: DASHBOARD GENERATION

Dashboard created successfully.
Dashboard file: outputs\customer_sales_dashboard.html

Step 13: Final Report

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STEP 13: FINAL REPORT
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Final report saved to: outputs\customer_sales_report.txt

PRACTICAL 12 COMPLETED SUCCESSFULLY.
Open this file in your browser:
outputs\customer_sales_dashboard.html
```

Result

The customer sales dashboard was implemented using the generated Superstore-style sales dataset. The workflow includes data loading, validation, KPI calculation, regional and category analysis, time-based analysis, product profitability, customer segmentation, interactive filter preparation and dashboard generation.

The dataset contains 10,000 records. The generated dataset produced Total Sales of \$19,856,617.22, Total Profit of \$2,721,446.80, and 10,000 unique orders. The overall profit margin is 13.71%.

Key Questions / Analysis

Q1. What is the role of interactive dashboards in business analytics?

Interactive dashboards convert large amounts of business data into visual and easy-to-explore information. They allow users to monitor KPIs, compare performance and investigate trends without manually preparing separate reports.

Q2. How do filters improve visual exploration?

Filters allow users to focus on selected regions, categories, customer segments or time periods. This reduces unnecessary information and helps users compare specific business conditions quickly.

Q3. What are KPI indicators?

Key Performance Indicators are measurable values used to monitor business performance. In this practical, Total Sales, Total Profit, Total Orders, Profit Margin and Average Order Value are used as important indicators.

Q4. Why are dashboards preferred over static reports?

Dashboards provide interactive, visual and frequently updateable views of business information. Unlike static reports, users can filter, compare and explore the data according to their needs.

Q5. How does user-driven exploration support decision-making?

User-driven exploration allows management to investigate specific regions, products, categories and time periods. This supports faster identification of high-performing areas, weak performance and opportunities for improvement.

Supplementary Problems

- Add customer segmentation analysis.
- Create a product-wise profitability dashboard.
- Compare quarterly sales performance.
- Implement drill-through pages or equivalent detailed views.
- Create an executive summary dashboard.
- Design an additional KPI to monitor customer retention.

Tools / Technology used

Python | Pandas | Plotly | CSV Dataset | Tableau / Power BI for dashboard visualization

Conclusion

The customer sales dashboard demonstrates how interactive analytics can support retail business decision-making. The dataset was cleaned and validated, KPIs were calculated, and sales were analyzed across regions, categories, products, customer segments and time periods. Interactive filters and visual summaries make it easier for management to explore performance and identify business opportunities.

Github Link:

https://github.com/RudraMaiyariya/D25DIT081_BDA_SHP_2026/tree/main/Practical12

